

PYTHON SPECIFIC HANDBOOK

Python Programming Quizzes & Starter Projects

*A targeted guide containing Python core concepts, code output quizzes,
and interactive projects extracted from the reels feed.*

Prepared for: Python Developers

Generated automatically from source video content.

1. Python Programming Quizzes

This section compiles the code output challenges from the reels feed. Practice answering these to improve your understanding of Python datatypes, operations, and control structures.

Quiz: 1 (python_quiz)

```
x = {"a": 1, "b": 2}
print(x["b"])
```

A) 1 B) 2 C) "b" D) Error

Correct Option: B (2)

Quiz: 2 (Day_69_Python_quiz_Comm)

```
x = [1, 2, 3]
x.append([4, 5])
print(len(x))
```

A) 3 B) 4 C) 5 D) Error

Correct Option: B (4) -- Note: append adds the list as a single element [4, 5] at index 3.

Quiz: 3 (Comment_your_answer_1)

```
print(0.1 + 0.2 == 0.3)
```

A) True B) False C) Error D) 0.3

Correct Option: B (False) -- Note: Due to floating-point precision in computers, 0.1 + 0.2 is actually 0.30000000000000004.

Quiz: 4 (Comment_your_answer_2)

```
x = 10
if x > 5:
    if x < 8:
        print("A")
    else:
        if x == 10:
            print("B")
        else:
            print("C")
else:
    print("D")
```

A) A B) B C) C D) D

Correct Option: B -- Note: x is 10, which is > 5. Since x is not < 8, it enters the outer else block. Since x == 10, it prints 'B'.

Quiz: 5 (Comment_your_answer_3)

```
x = {1, 2, 3}
y = {3, 4, 5}
print(x & y)
```

A) {1, 2, 3, 4, 5} B) {3} C) {1, 2} D) set()

Correct Option: B ({3}) -- Note: The '&' operator performs set intersection, returning elements common to both sets.

Quiz: 6 (Python_Quiz_Answers)

```
x = [1, 2, 3]
print(x + [4, 5])
```

A) [1, 2, 3, 4, 5] B) [5, 7, 8] C) [[1, 2, 3], [4, 5]] D) Error

Correct Option: A ([1, 2, 3, 4, 5]) -- Note: Adding lists in Python performs list concatenation.

2. Python Starter Projects

Below are the utility and graphics scripts demonstrated in the "Day X of learning Python" video tutorial series. You can run these scripts directly on your local Python environment.

Basic Login System (Day 8)

A simple Python script to validate a user login. It takes a username and password from standard input and compares them against preset hardcoded variables.

```
user = "admin"
password = "3690"
u = input("Username: ")
p = input("Password: ")
if u == user and p == password:
    print("Login Successful !")
else:
    print("Login Failed !")
```

Open Website Automatically (Day 35)

An automation script that counts down 3 seconds, prints a message, and opens a web URL in the default system web browser.

```
import webbrowser
import time

print("Opening website in 3 seconds...")
time.sleep(3)
url = "https://www.flipkart.com/"
webbrowser.open(url)
```

Even / Odd Checker (Day 69)

Takes an integer input from the user and checks if the remainder after division by 2 is zero. Prints 'Even' or 'Odd' accordingly.

```
num = int(input("Enter number: "))
if num % 2 == 0:
    result = "Even"
else:
    result = "Odd"
print("Number:", num)
print("Result:", result)
print("Done!")
```

Random Username Generator (Day 70)

Generates a random username by concatenating a random selection from a list of prefixes with a random 3-digit number.

```
import random

names = ["Cyber", "Code", "Tech", "Python"]
num = random.randint(100, 999)
username = random.choice(names) + str(num)
print("Generated Username:")
print(username)
print("Done!")
```

Turtle Graphics Color Spiral (Day 72)

Uses Python's turtle graphics module to draw an interactive color spiral, rotating through red, green, and blue lines with increasing forward distance.

```
import turtle

t = turtle.Turtle("circle")
t.speed(67)
cs = ['red', 'green', 'blue']
for i in range(0, 300, 2):
    t.color(cs[i % len(cs)])
    t.left(90)
    t.forward(20 + i)
```

Console Quiz Game (Day 71)

An interactive console application that asks the user a multiple choice question about the creator of Python, checks the answer, and increments their score.

```
score = 0
print("=== Quiz Game ===")
print("1) Who is the creater of python?")
print("A) Dennie Ratchie")
print("B) Bjarne Strustrup")
print("C) Guido Van Rossum")
print("D) Jamps Gosling")
answer = input("Enter your choice(A/B/C/D): ")
if answer == "C" or answer == "c":
    print("Correct! well done .")
    score = score + 1
else:
    print("Worng! The correct answer is C) \n (\"Guido Van\")")
```